

PRINCETON UNIVERSITY

ZERO LAB

Zero-carbon Energy Systems Research and Optimization Laboratory

Meditations on power system planning under uncertainty

Jesse D. Jenkins

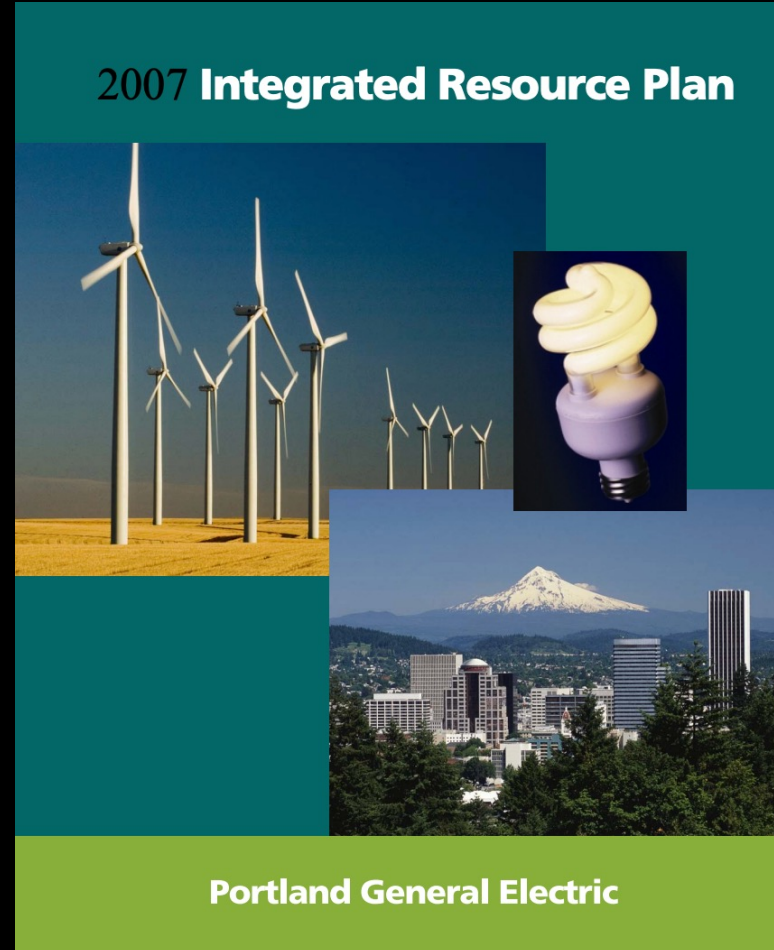
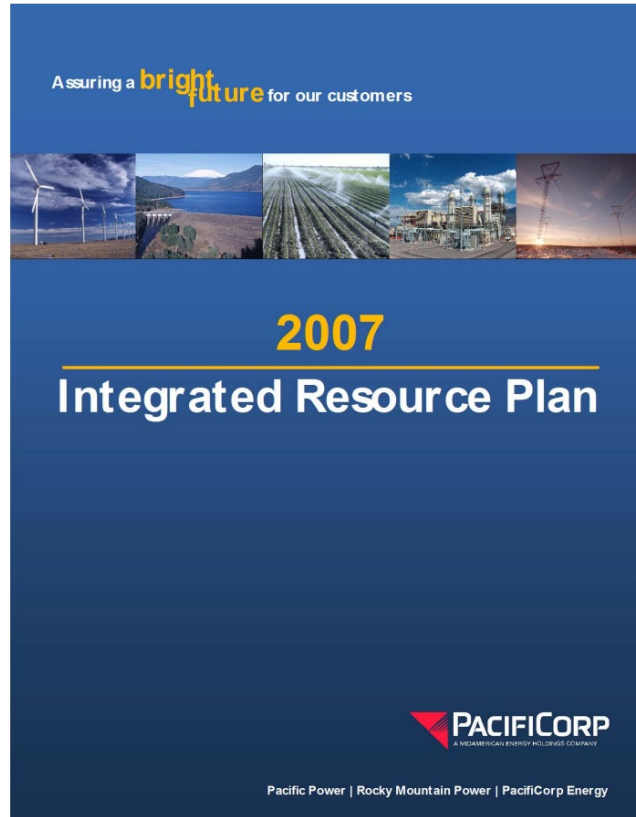
September 11, 2026 – PowerUp 2026 Workshop

This talk:

1. My journey starting with planning and risk
3. Why “best practices” still need to get better
4. Making robust planning tractable

1. My journey begins with planning and risk

My journey begins... with Integrated Resource Plans!



BEFORE THE PUBLIC UTILITY COMMISSION
OF OREGON

LC 42

 **RENEWABLE
NORTHWEST**

In the Matter of)
PUBLIC UTILITY COMMISSION OF)
OREGON,)
In the Matter of PACIFICORP, dba)
PACIFIC POWER & LIGHT COMPANY)
Application for Acknowledgement of its)
2007 Integrated Resource Plan.)

OPENING COMMENTS ON
PACIFICORP 2007 INTEGRATED
RESOURCE PLAN BY RENEWABLE
NORTHWEST PROJECT

Introduction

The Renewable Northwest Project (RNP) provides these opening comments on PacifiCorp's 2007 Integrated Resource Plan (IRP). While we have several concerns with this 2007 IRP, it is clearly an impressive document, both in its scope and its technical sophistication. RNP recognizes the difficulties in planning for the future amidst risk and uncertainty and appreciates the efforts PacifiCorp has put into developing this IRP.

We applaud PacifiCorp's inclusion of 2,000 megawatts of new renewable energy resources in the Action Plan and Preferred Portfolio and are pleased to see that the Company has selected a quantity of renewables that exceeds the commitments made by Mid-American Energy Holdings Company (MEHC) during their acquisition of PacifiCorp. This demonstrates that the addition of new renewable energy resources to PacifiCorp's resource portfolio stands on its own merits as a least-cost, least-risk strategy to reduce fuel price volatility, achieve fuel diversity, reduce risk of future environmental regulations and comply with existing and potential state and federal Renewable Portfolio Standards (RPS).

We do have some major concerns with this 2007 IRP, mainly centered on the continued presence of two new pulverized coal plants and the overall increase in carbon dioxide emissions resulting from PacifiCorp's preferred portfolio. We believe the proposed PacifiCorp IRP filing is flawed in the following manner:

“Best mix of cost and risk”



ORDER NO. 07-002

ENTERED 01/08/07

”..we expressed the goal that utility resource plans should identify resources that provide the best mix of cost and risk.

PUBLIC UTILITY COMMISSION OF)
OREGON)

ORDER

Investigation Into Integrated Resource)
Planning.)

DISPOSITION: GUIDELINES ADOPTED; RULEMAKING
AND INVESTIGATION OPENED

The Public Utility Commission of Oregon (Commission) opened this docket to investigate whether the requirements for least-cost planning, first established in Order No. 89-507, should be revised. Following a suspension of proceedings pending decisions in other active dockets,¹ we issued Order No. 05-133, which instructed the parties in this docket to “focus on cost, not market.” In so doing, we expressed a goal that utility resource plans should identify resources that provide the best mix of cost and risk.

“Best mix of cost and risk”



b. Risk and uncertainty must be considered.

- *At a minimum, utilities should address the following sources of risk and uncertainty:*
 1. *Electric utilities: load requirements, hydroelectric generation, plant forced outages, fuel prices, electricity prices, and costs to comply with any regulation of greenhouse gas emissions.*
 2. *Natural gas utilities: demand (peak, swing and base-load), commodity supply and price, transportation availability and price, and costs to comply with any regulation of greenhouse gas emissions.*
- *Utilities should identify in their plans any additional sources of risk and uncertainty.*

c. The primary goal must be the selection of a portfolio of resources with the best combination of expected costs and associated risks and uncertainties for the utility and its customers.⁶

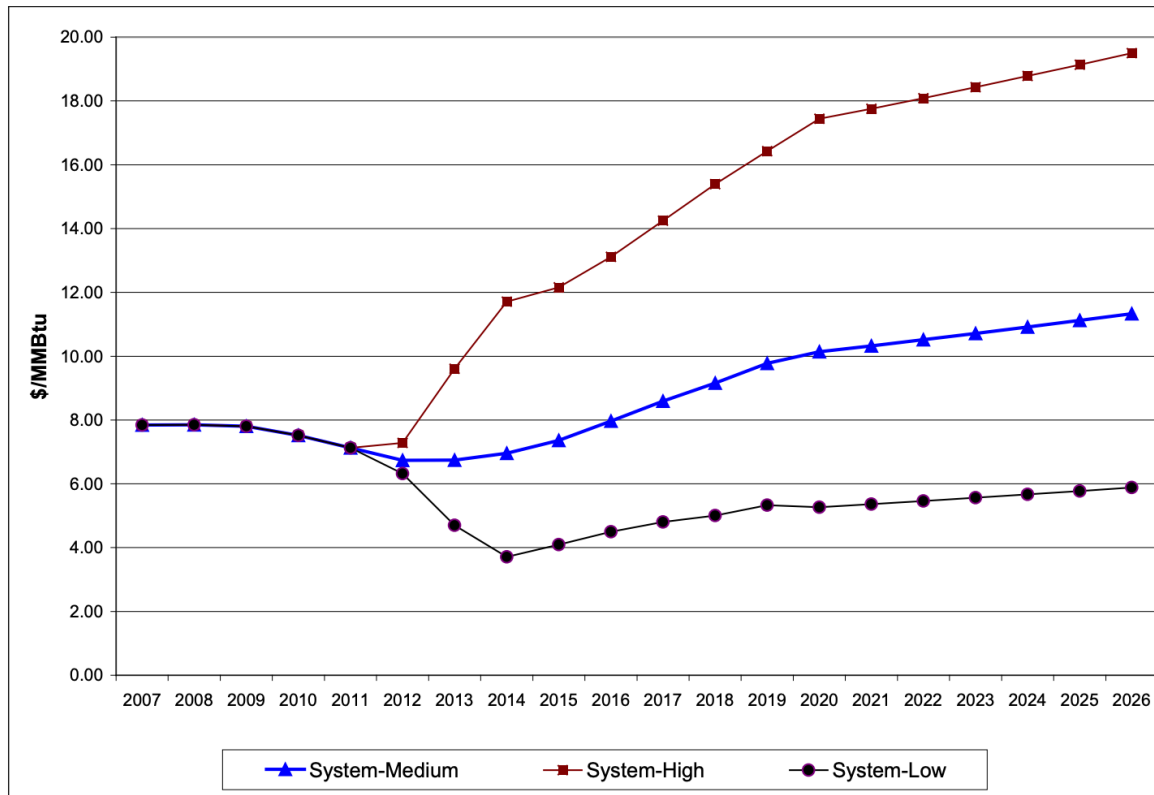
What does least cost really mean?



”At the outset, ICNU prefers the continued use of “least cost” over “the best combination of expected costs and associated risks” to describe the desired selection of a portfolio of resources. ICNU Reply Comments at 6. ICNU explains that such focus has placed too much emphasis on risks that have not materialized, such as CO2 regulatory costs. We disagree. The goal of the IRP is to help identify the lowest realized cost over the planning horizon. To accomplish this, risks must be considered, as one resource decision with the lowest expected cost might, due to associated risks, result in higher costs than other resource strategies.

Grappling with risk

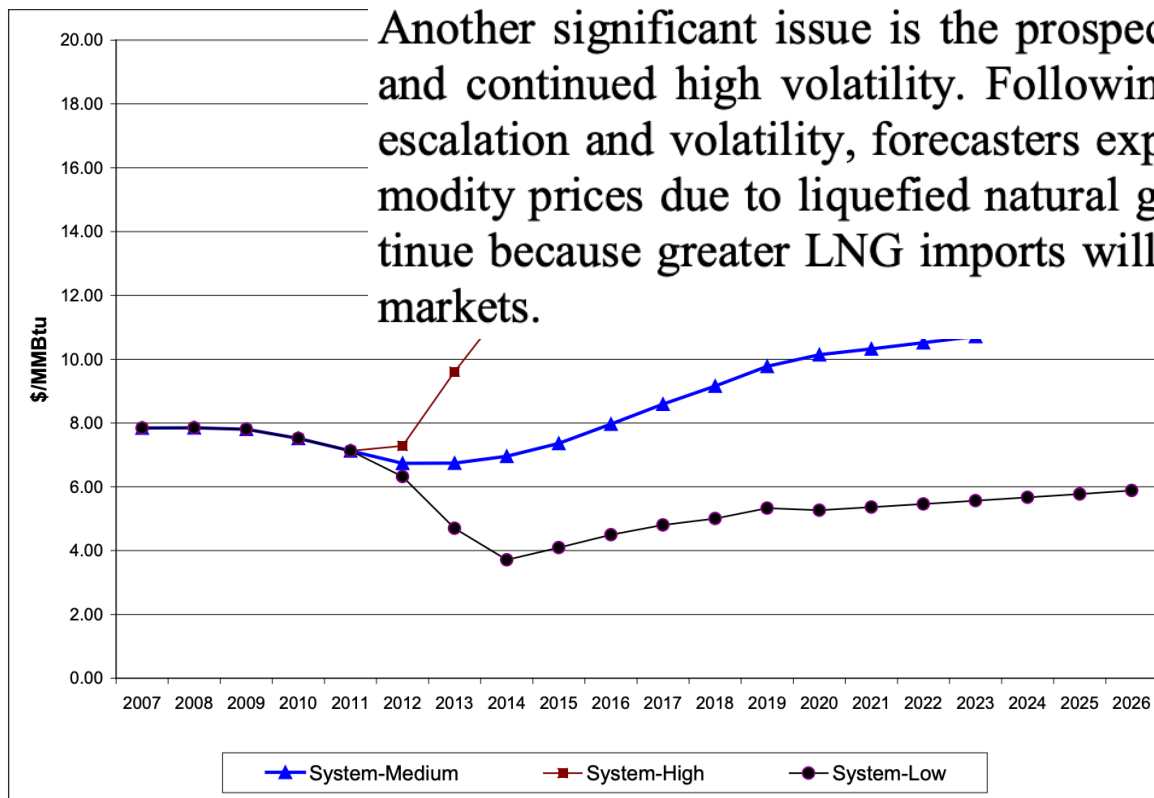
Figure 6.2 – System Average Annual Natural Gas Prices: Low, Medium, and High Scenario Values



Source: PacifiCorp 2007 IRP

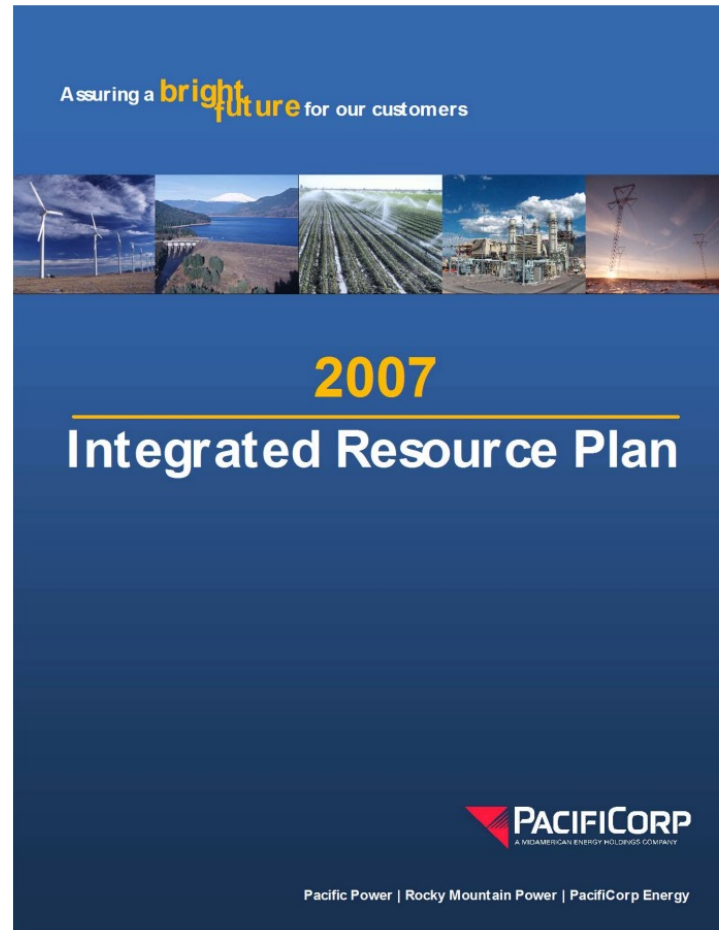
Grappling with risk

Figure 6.2 – System Average Annual Natural Gas Prices: Low, Medium, and High Scenario Values

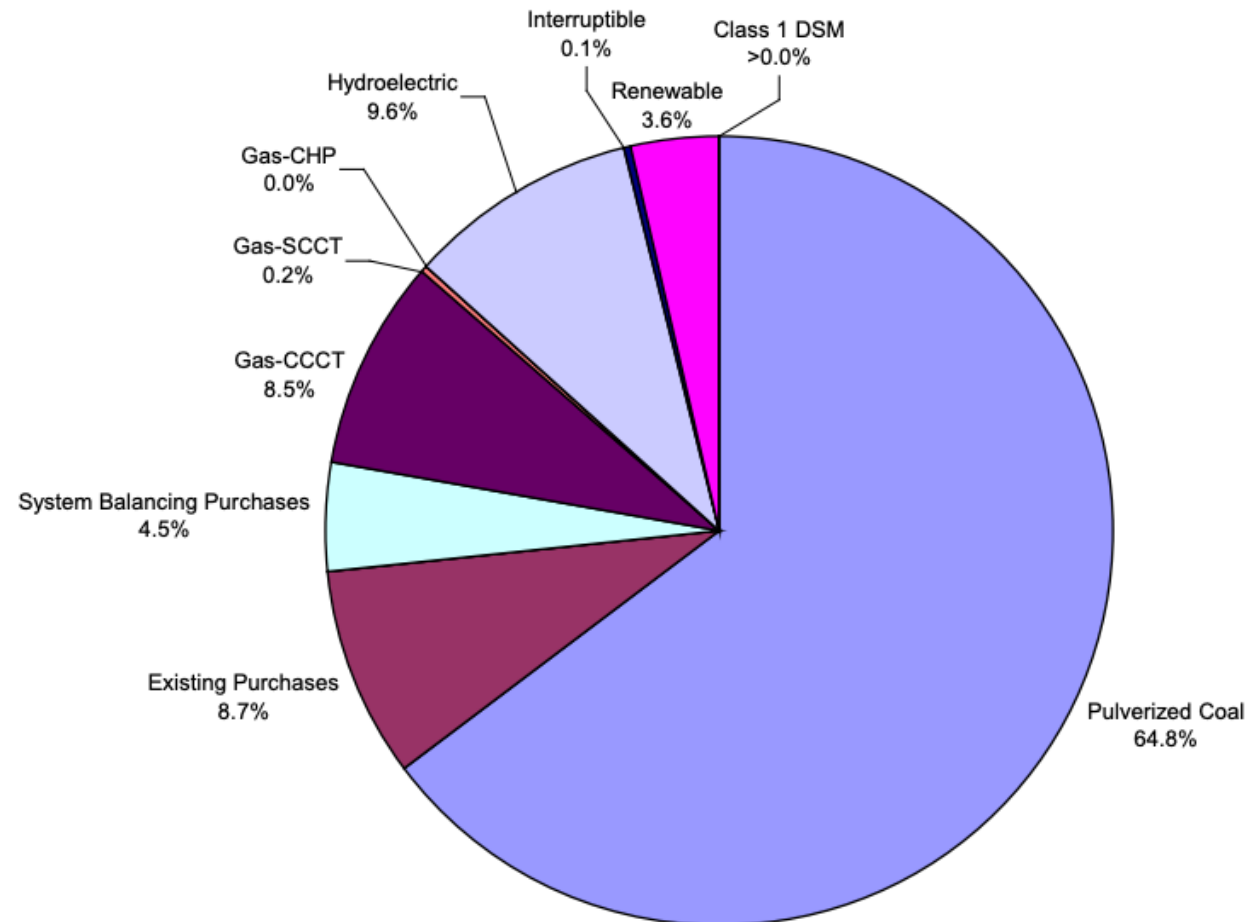


Source: PacifiCorp 2007 IRP

Build more coal? What could go wrong...



2007 Resource Energy Mix with Preferred Portfolio Resources
(Average for five CO2 Adder Cases)



Grappling with risk

One of the largest issues emerging from governmental policy and regulatory initiatives is how to plan given an eventual, but highly uncertain, climate change regulatory regime. Not only have there been significant policy developments for currently-regulated pollutants, but there have also been important state-level climate change regulatory initiatives. Other regulatory issues include state renewable portfolio standards, hydropower relicensing, and major relevant provisions of the Energy Policy Act of 2005.

Source: PacifiCorp 2007 IRP

A balanced plan?

“The 2,629 MW of thermal generation capacity consists of four thermal units in the east (two fueled with coal and two with natural gas) and one natural gas unit in the west.

The most robust resource strategy relies on total resources creating a diverse portfolio of resources including renewables and demand side management combined with natural gas and coal-fired generating resources.

Two major issues hang over the most significant resource choices that PacifiCorp must make (i) the future cost of natural gas and (ii) the future cost of or constraints on air emissions, and carbon dioxide emissions in particular. PacifiCorp believes it has adequately addressed these risks in the analysis, based on our current understanding of these issues.

Source: PacifiCorp 2007 IRP

The role of stakeholders in robust planning

“E. The IRP Fails to Adequately Plan for Future CO2 Regulation

The IRP enunciates a “Corporate Greenhouse Gas Mitigation Strategy” that not only seems insufficient to the task, but is hardly an active strategy at all. PacifiCorp’s general ‘mitigation strategy’ seems to be to wait until there are changes in the law... While the Company includes more emissions-free resources and demand side management in the Preferred Portfolio than in earlier candidate portfolios (i.e. Group 1 portfolios), these efforts are insufficient to reduce the Company’s total greenhouse gas emissions when the Portfolio also includes additional new carbon-intensive resources, including pulverized coal.

We believe the Company must reduce its total greenhouse gas emissions. Any plan that results in an increase in greenhouse gas emissions can no longer be considered prudent, given the current scientific certainty and regulatory environment on global climate change. We contend that any proposal to add carbon-intensive power facilities could only be acceptable in the context of an overall plan to reduce total emissions related to serving load.

The Commission's decision



Commission Disposition

“We adopt Staff’s recommended exceptions to the Action Plan [modifying Pacific Power’s proposed thermal resource procurements to exclude new pulverized coal and revisit baseload vs peaking requirements].

(ORDER NO. 08-232)

Policy to shape planning

74th OREGON LEGISLATIVE ASSEMBLY--2007 Regular Session

Enrolled
Senate Bill 838

Sponsored by Senator AVAKIAN; Senators ATKINSON, BATES, BROWN, BURDICK, CARTER, COURTNEY, DEVLIN, GORDLY, METSGER, MONNES ANDERSON, MONROE, MORRISETTE, PROZANSKI, STARR, WALKER, WESTLUND, Representatives DINGFELDER, ROSENBAUM (at the request of Governor Theodore R. Kulongoski)

CHAPTER

AN ACT

Relating to electricity; creating new provisions; amending ORS 261.010, 261.015, 261.020, 261.250, 261.253, 261.305, 261.335, 261.348, 261.355, 262.005, 262.015, 262.075, and declaring an emergency.

Whereas the Legislative Assembly finds that it is in the interest of the search and development of new renewable energy sources in Oregon; and

Whereas the Legislative Assembly finds that it is necessary for Oregon's decrease their reliance on fossil fuels for electricity generation and to increase their use of renewable energy sources; and

Whereas this 2007 Act may be cited as the Oregon Renewable Energy Act; and

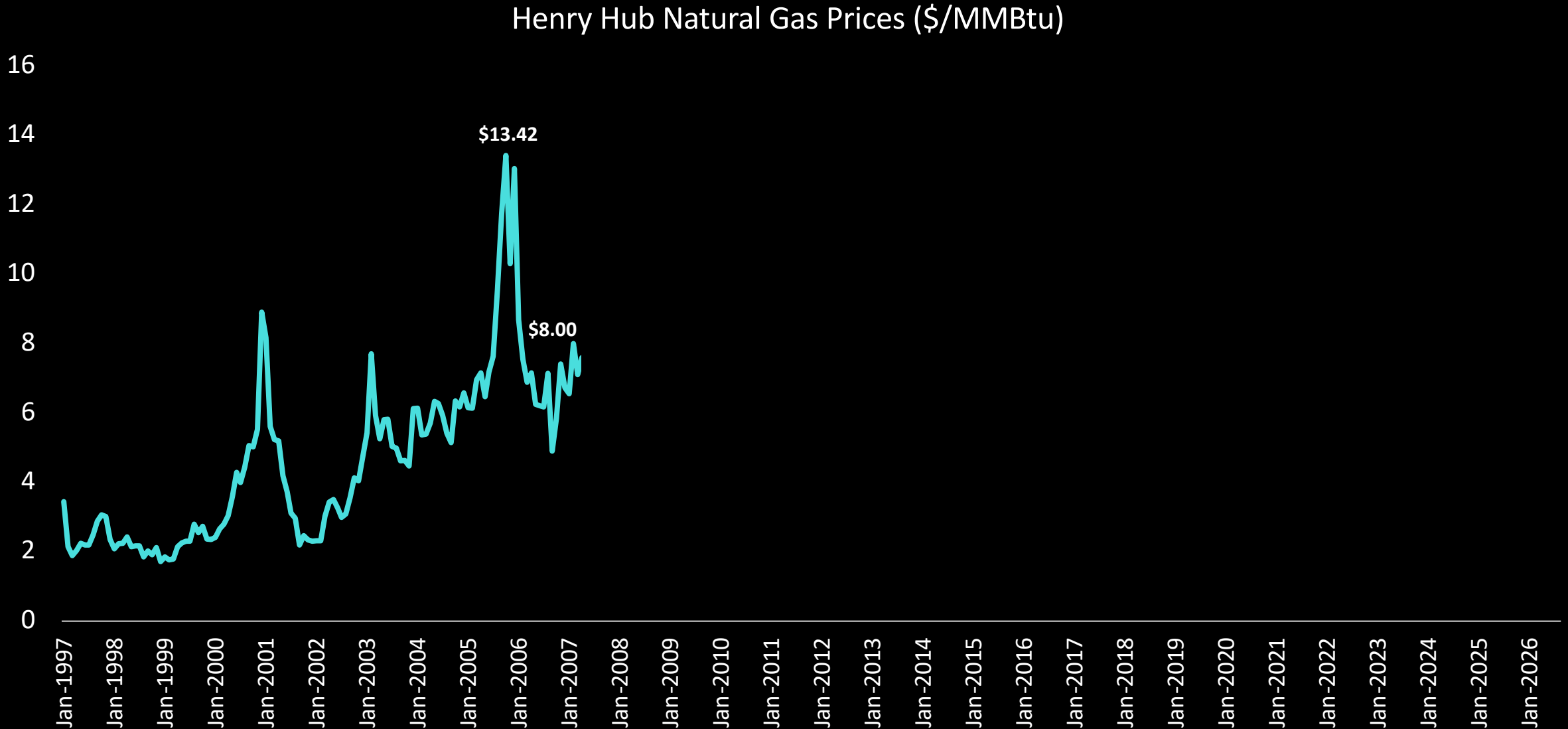
Whereas the Oregon Renewable Energy Act provides a comprehensive renewable energy policy for Oregon, enabling industry, government and all Oregonians to accelerate the transition to a more reliable and more affordable energy system; now, therefore,

Be It Enacted by the People of the State of Oregon:

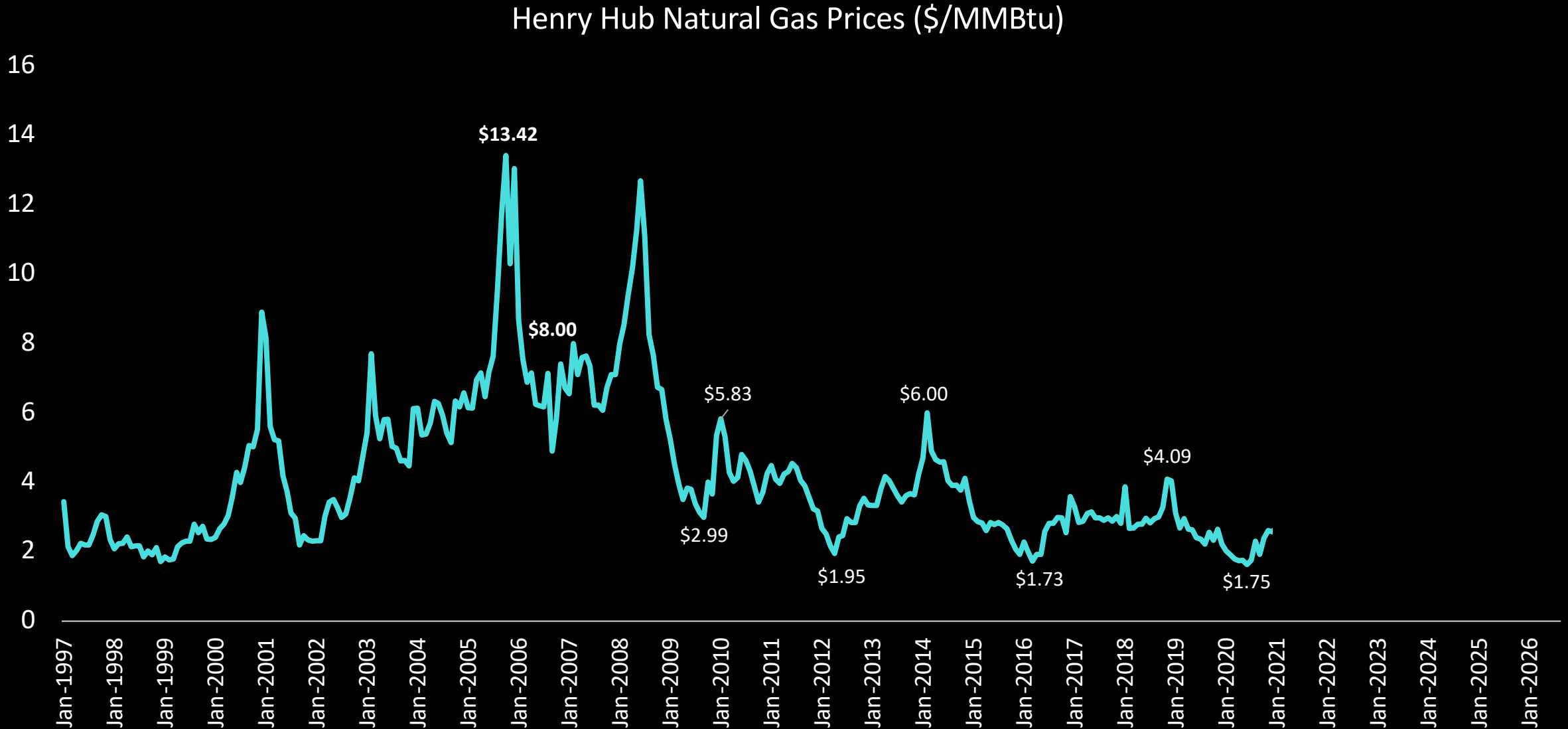
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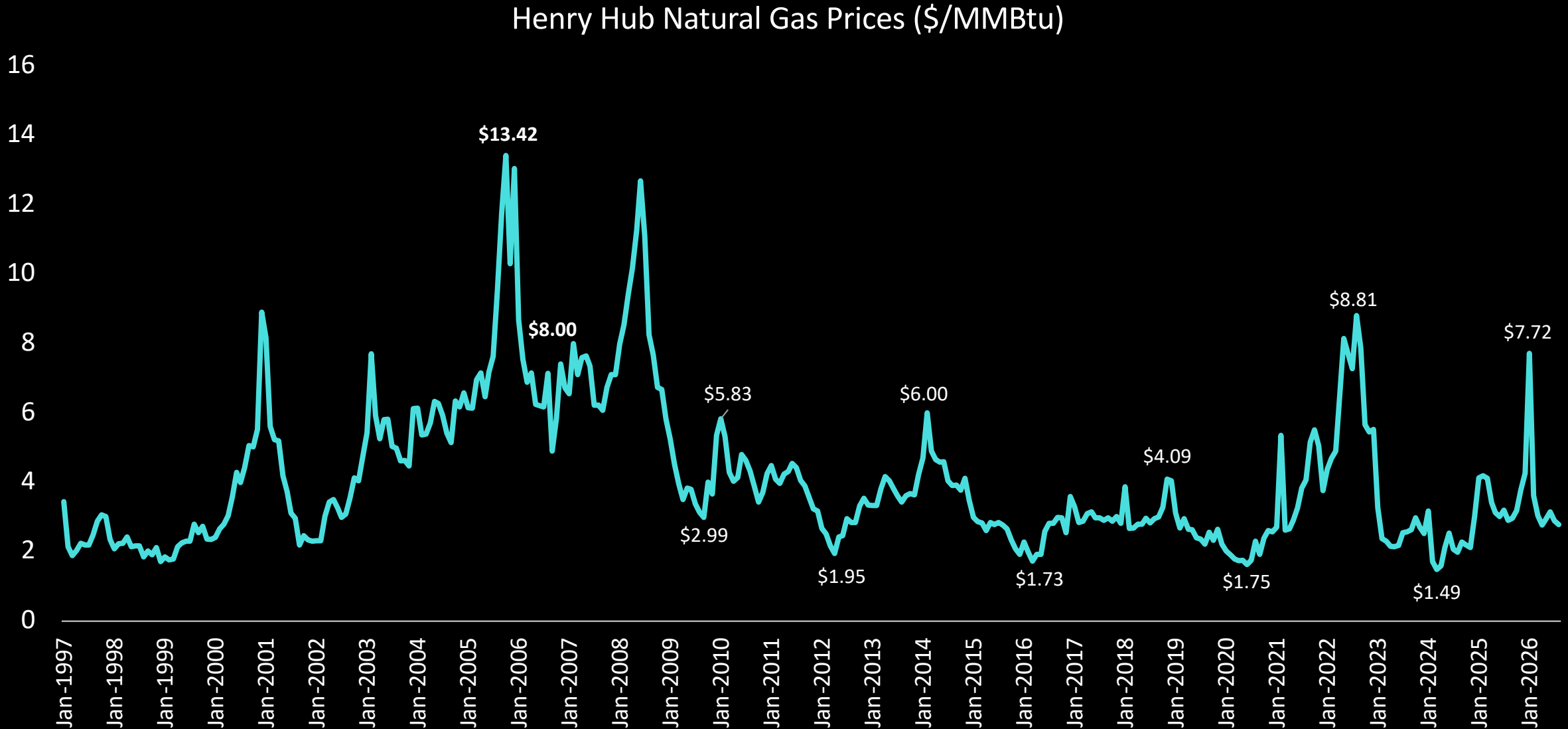
The risks (almost) no one saw coming...



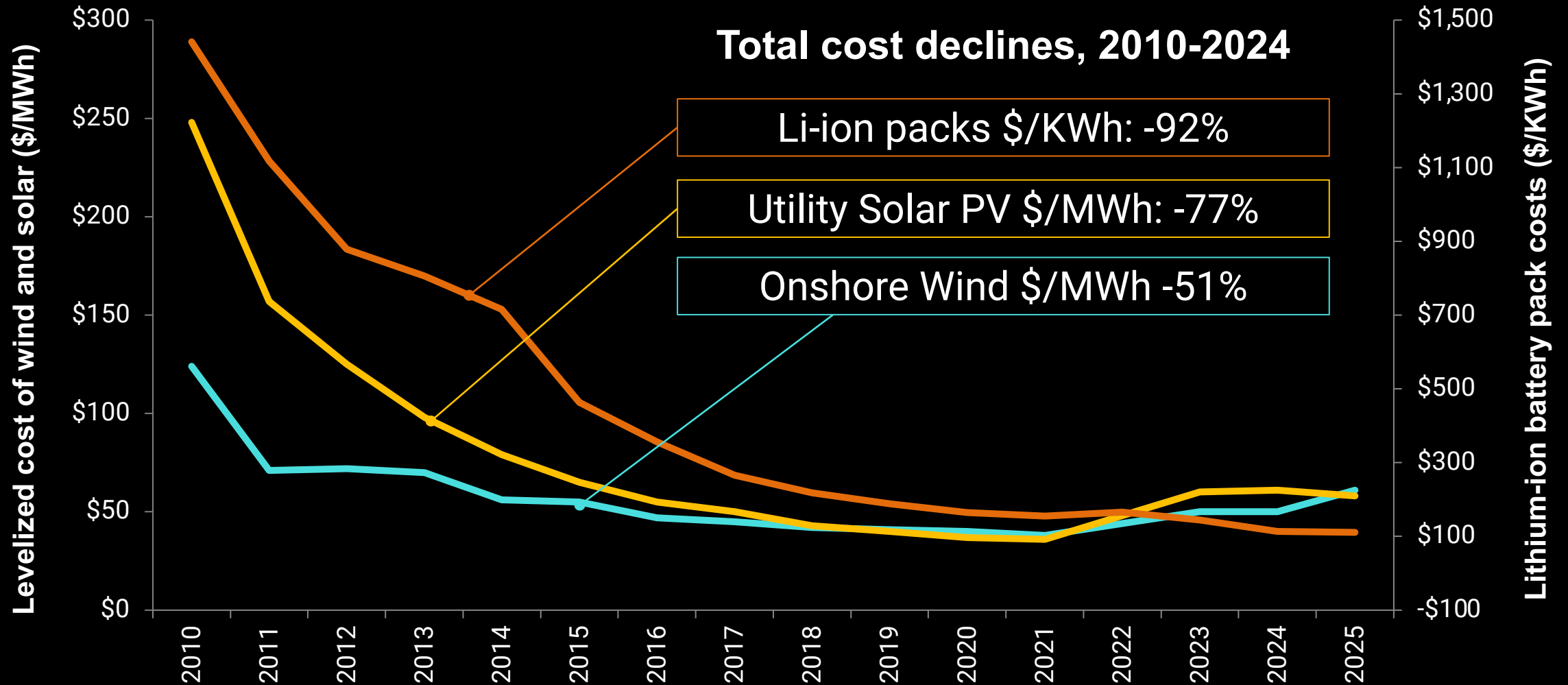
The risks (almost) no one saw coming...



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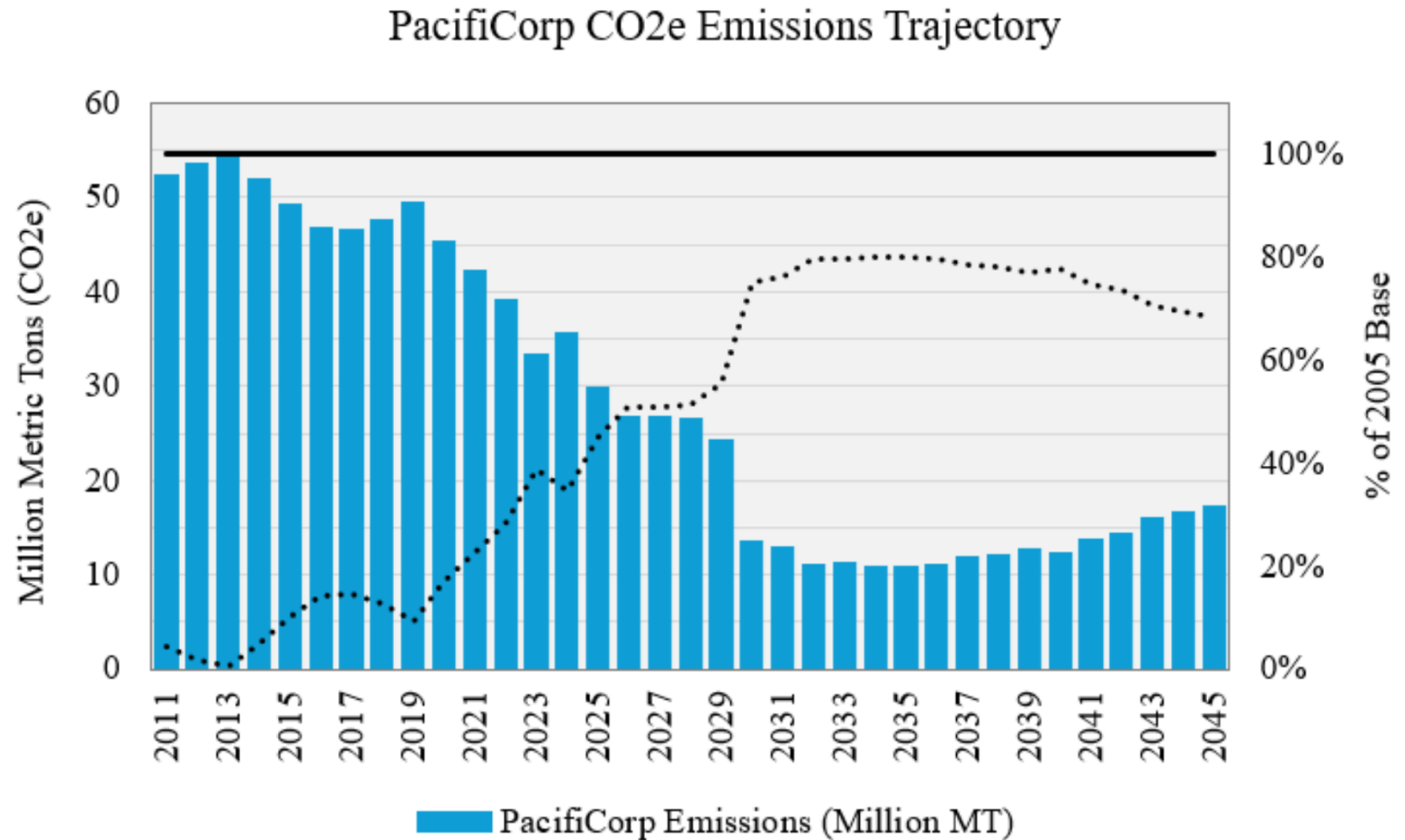


Data Sources: Wind & solar costs from Lazard (2023), 2023 Levelized Cost of Energy+.
Battery pack costs from Bloomberg New Energy Finance (2023), Battery Price Survey.

Where we ended up

2025 PacifiCorp IRP preferred portfolio:

- **3.7 GW new wind**
- **7.5 GW new battery storage**
- **5.9 GW new solar**
- **500 MW new nuclear (Sodium advanced reactor)**
- **5.2 GW energy efficiency savings**
- **0.8 GW direct load control programs**
- **Exit from Colstrip coal plant in MT by 2030**
- **Coal to gas conversions: Naughton (2026) & Dave Johnson (2029) in WY**
- **Evaluate CCS for Jim Bridger in WY by 2030**



2. Why "best practices" still need to get better

What is “best practice” then and now?

“For resource screening, the company used the CEM to evaluate generation, load control, price-responsive demand-side management, market purchases, and transmission resources on a comparable basis with the use of “alternative future” scenarios.

The main purpose of these scenarios is to identify general resource patterns attributable to changes in assumptions, and to help identify robust resources—those that frequently appear in the model’s optimized portfolios under a range of futures. ...

Using the results from the alternative future scenario studies, PacifiCorp defined risk analysis portfolios for stochastic simulation. The CEM was used to help build fixed resource investment schedules... The resulting portfolios were then simulated using the PaR model.

The PaR simulations incorporate stochastic risk in its production cost estimates by using Monte Carlo random sampling of five stochastic variables: loads, commodity natural gas prices, wholesale power prices, hydro energy availability, and thermal unit availability.

What is “best practice” then and now?



“**Order No. 1920** stated that transmission providers must properly account for known causes of long-term transmission needs to ensure just and reasonable rates. To do so, **transmission providers must use scenarios** to identify long-term transmission needs and transmission facilities to meet those needs. In long-term planning, transmission providers must develop **a minimum of three distinct long-term scenarios** using best available data inputs and **no less than a 20-year transmission planning horizon**, and must reassess the scenarios at least once every five years.

Summary of FERC Order 1920, May 2024

Source: <https://www.ferc.gov/explainer-transmission-planning-and-cost-allocation-final-rule>

Two classes of uncertainty

1. Stochastic Uncertainties

“One of the principle sources of risk that is addressed in this IRP is **stochastic risk**. Stochastic risks are **quantifiable uncertainties for particular variables**. The variables addressed in this IRP include retail loads, natural gas prices, wholesale electricity prices, hydroelectric generation, and thermal unit availability.

2. Deep Uncertainties

“PacifiCorp defines a **scenario risk as an externally-driven fundamental and persistent change** to the expected value of some parameter that is expected to significantly impact portfolio costs. This risk category is **intended to embrace abrupt changes to risk factors that are not amenable to stochastic analysis**.

Two classes of uncertainty

Examples of Stochastic Uncertainties in Grid Planning

- **Weather-related** parameters:
 - demand, wind & solar profiles
 - hydrology
 - thermal plant derates/outage risks
- **Forced plant outages**
- **Short-term fuel price** fluctuations?
- **Short-term market price** volatility?

Examples of Deep Uncertainties in Grid Planning

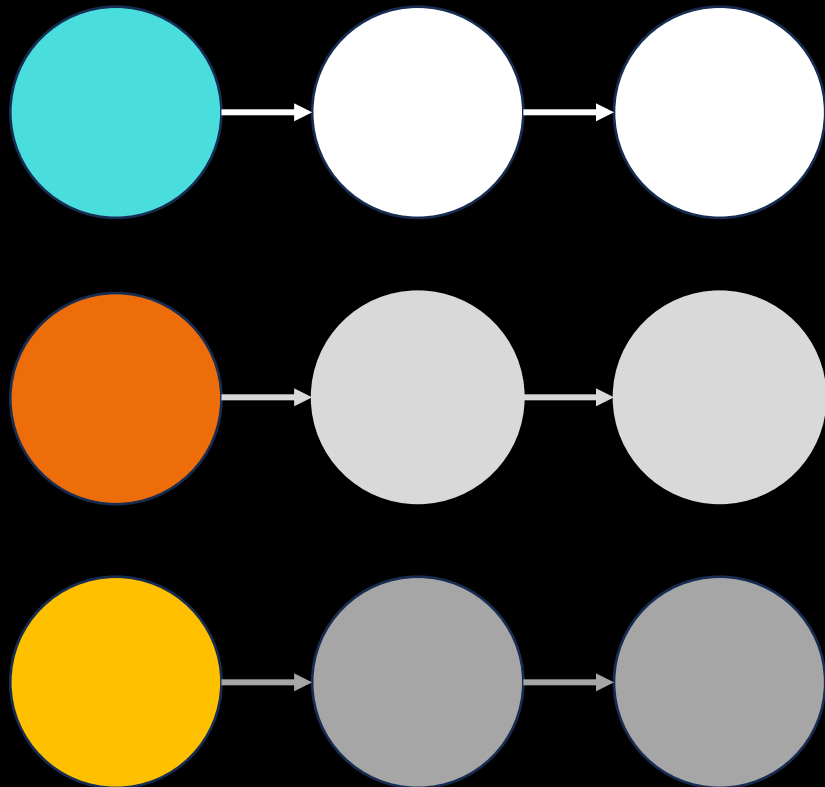
- Future changes to **policy and regulation**
- Future **technology costs**, especially for emerging technologies
- **Fuel price shocks** (e.g. geopolitical risks)
- **Changes in demand growth paradigm**, including electrification & data center loads, macro-economic conditions
- **Siting constraints** for renewables, nuclear, transmission, etc. (e.g., social license)
- Many more...

Deterministic scenarios + Monte Carlo stress test

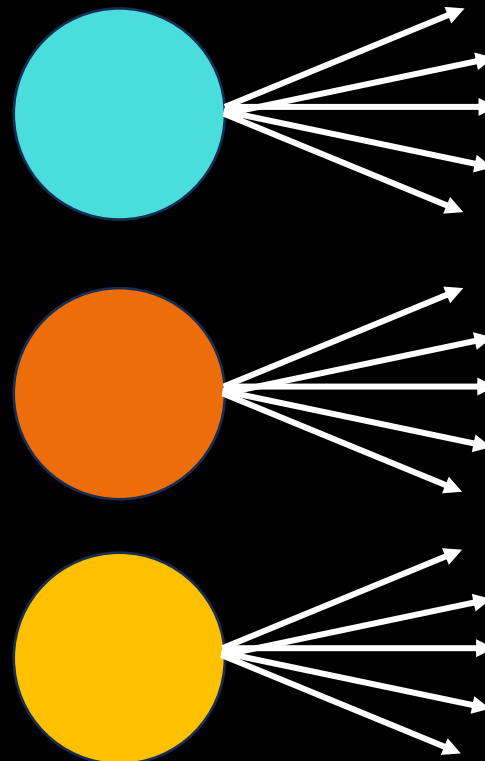
1. Deterministic Plan for Each Scenario

“Here & now”
decisions

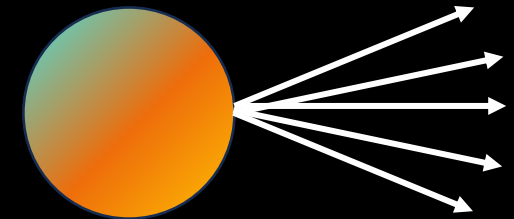
Deferred decisions



2. Stochastic Production Cost Simulation

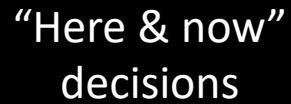


3. Preferred plan including “robust” capacity choices

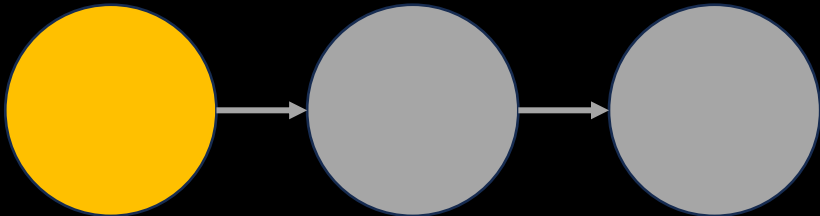
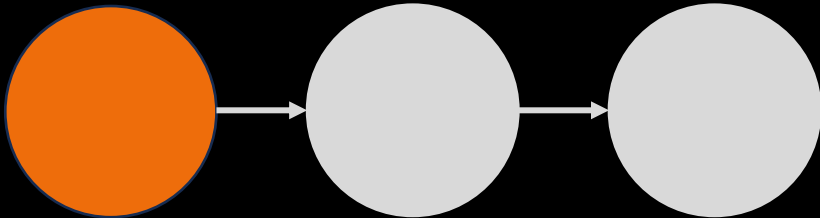
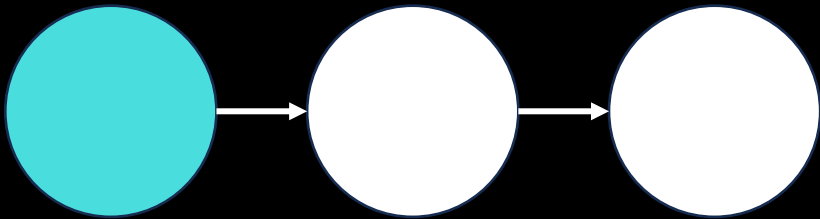


What's wrong with deterministic scenario planning?

1. Deterministic Plan for Each Scenario



Deferred decisions



“Here & Now” Decision	Scenario A: Cheap Gas, Expensive Coal	Scenario B: Cheap Coal, Expensive Gas
Build CCGT	\$5 billion	\$10 billion
Build Coal	\$10 billion	\$5 billion
Build Renewables & Storage	\$8 billion	\$8 billion

Example adapted from: Mantegna & Go 2026, “Integrated Resource Planning for an Uncertain Future A practitioner’s guide to managing uncertainty in electric sector resource planning, <https://www.resilient.energy/whitepaper>

Summary: Deterministic Scenario Planning

Pros

Simple to implement; can use existing models; computational simplicity

Helps generate a range of near-term decisions via deterministic scenarios

Cons

Portfolios that are flexible and adaptable may be missed, as the model has no reason to select them for any one scenario

Can have the appearance of sufficing for an uncertainty management framework in and of itself

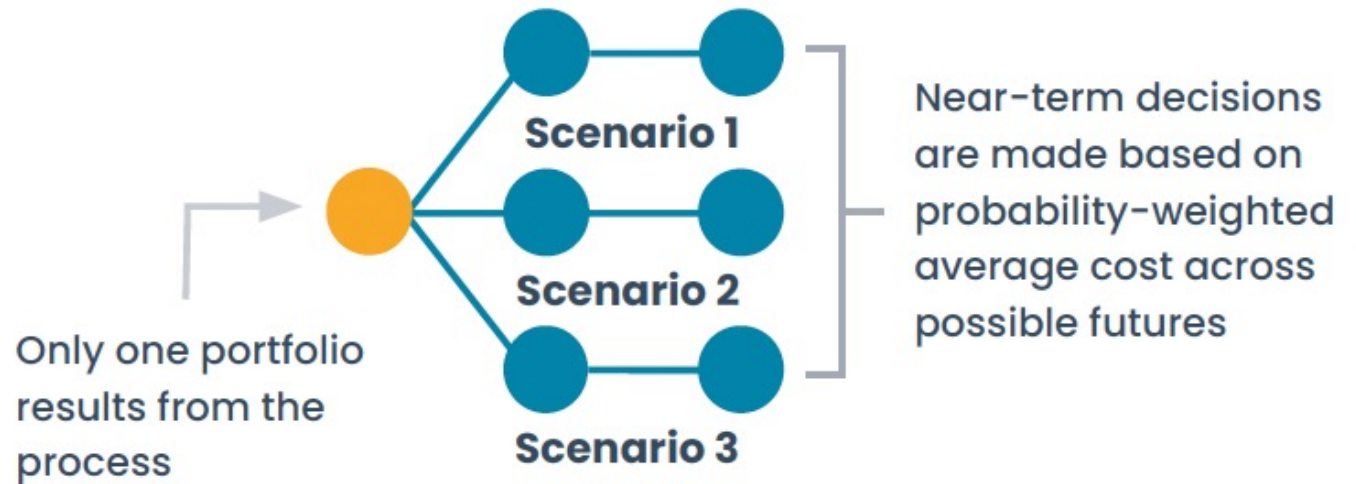
Source: Mantegna & Go 2026, “Integrated Resource Planning for an Uncertain Future A practitioner’s guide to managing uncertainty in electric sector resource planning, <https://www.resilient.energy/whitepaper>

Contemporary examples:

- **PacifiCorp** 2025 Integrated Resource Plan
- **MISO** regional transmission planning
- **ENTSO-E** long-range transmission plan
- **FERC Order 1920**: Regional transmission plans must evaluate at least three diverse, plausible long-term scenarios factoring in changes in the power mix, demand growth, and extreme weather

Stochastic Optimization

FIGURE 2. Graphical illustration of stochastic optimization.



Pros

Can help find portfolios that are flexible, adaptable, and robust amidst a large decision space

Can incorporate probabilistic information where available

Cons

Requires subjective assignment of probabilities

More complex methodology; may be more difficult for stakeholders to understand; higher computational complexity

Who wants to place a bet?



?

Robust Optimization

FIGURE 3. Illustrative schematic of robust optimization in the context of resource planning.



Source: Mantegna & Go 2026, "Integrated Resource Planning for an Uncertain Future A practitioner's guide to managing uncertainty in electric sector resource planning, <https://www.resilient.energy/whitepaper>

What do we want to “insure” against?



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Robust Optimization

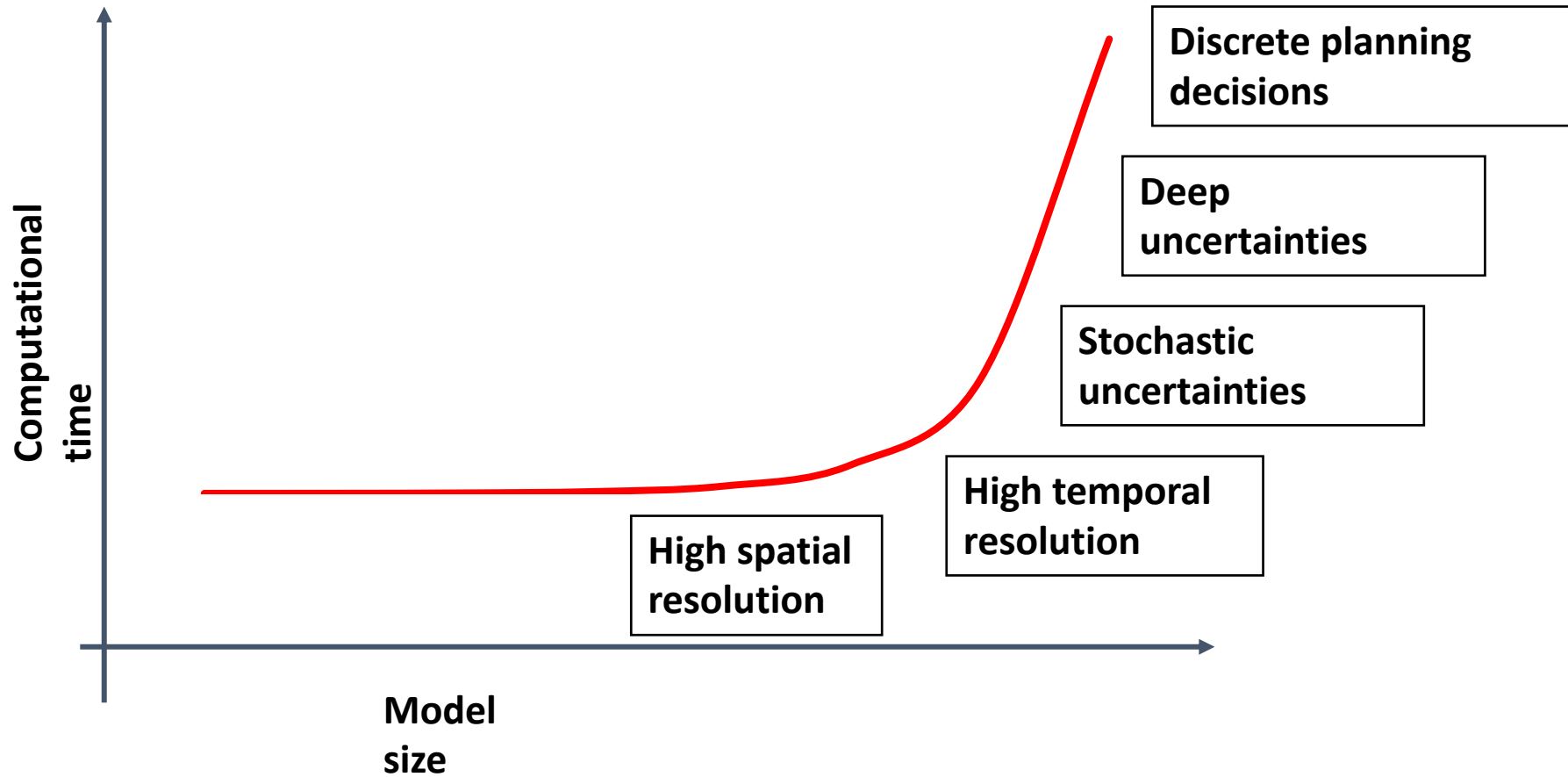
“[Robust optimization] is not the same as taking the applicable worst-case scenario, even if it is known, and optimizing for that scenario deterministically. This is because if one were to optimize for that scenario, the model would be likely to “over-optimize” to that scenario, thus potentially creating a new worst-case scenario from different parameters going wrong. Therefore, robust optimization should be thought of as “finding a solution that does best in the worst case,” not just “optimizing for the worst case.”

Pros	Cons
Computationally advantageous over stochastic planning	Still requires subjective assignment of parameter ranges and level of conservativeness
No assignment of probabilities necessary	Can be more difficult for stakeholders to understand than stochastic planning

Source: Mantegna & Go 2026, “Integrated Resource Planning for an Uncertain Future A practitioner’s guide to managing uncertainty in electric sector resource planning, <https://www.resilient.energy/whitepaper>

3. Making robust planning computationally tractable

The challenge: computational constraints



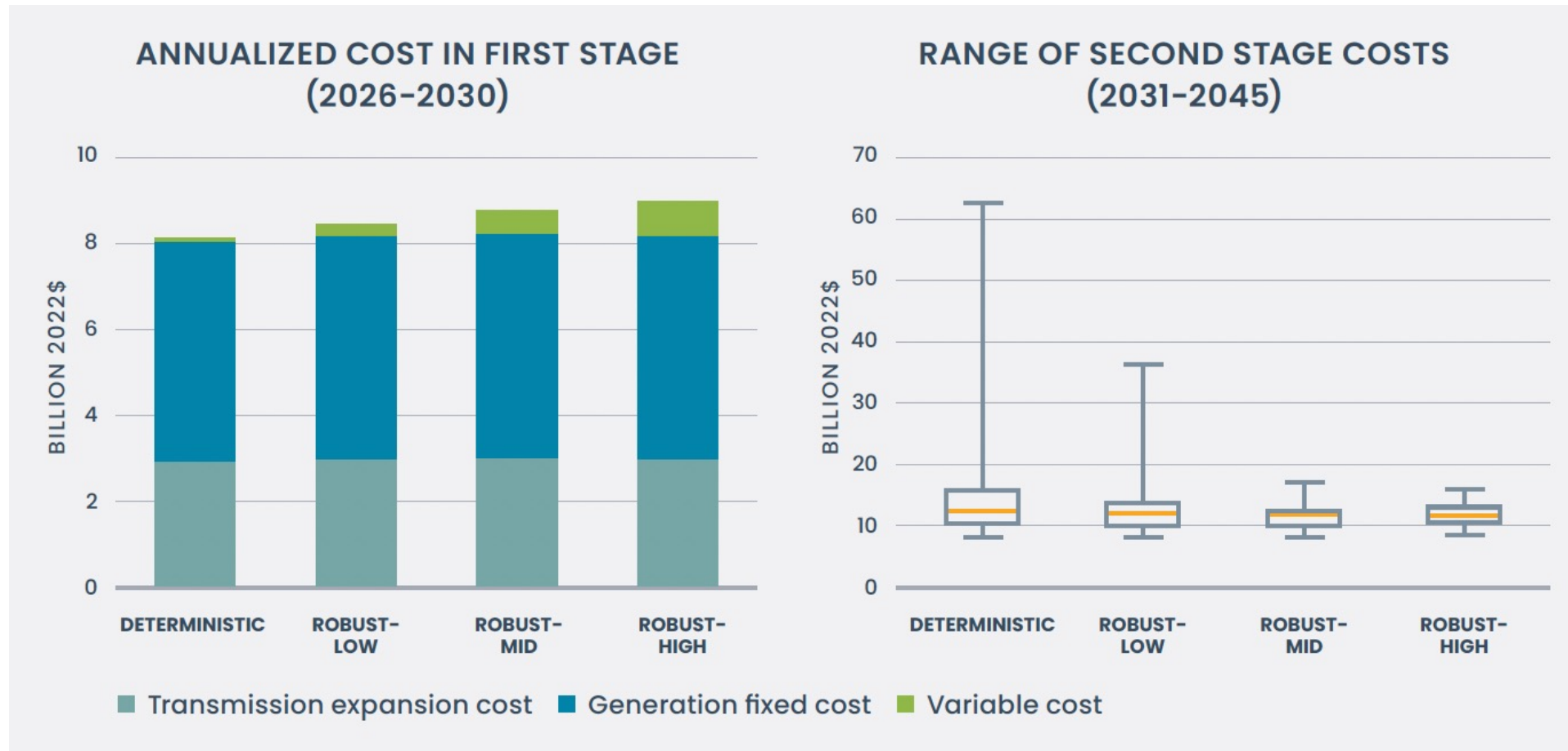
The “short blanket” theorem



Uncertainty-Aware Grid Planning in the Real World

A Method Enabling Large-Scale, Two-Stage Adaptive Robust Optimization for Capacity Expansion Planning

Gabriel Mantegna, Emil Dimanchev, Filippo Pecci, Neha Patankar, Jesse Jenkins (2026); <https://arxiv.org/abs/2603.00394>



Uncertainty-Aware Grid Planning in the Real World

How it works: “split budget” uncertainty sets

$$\min_{x \in \Psi} (c^T x + \max_{u \in U} \max_{s \in S} \min_{y_s \in \Omega_s(x)} b(u)^T y_s)$$

Near-term
(first stage)
planning decisions

Worst-case
cost across
uncertainty
set for **cost-
related**
uncertainties

Worst-case
cost across
set of all **RHS**
constraint
scenarios

Second-stage decisions for each scenario
and dualized cost uncertainty set

The “one weird trick”: we can solve a (large) LP if we use separate uncertainty sets for costs and all other uncertainties

New California Legislation: AB 2111

[Bills](#)

Share 

AB 2111: Electricity: transmission planning and transmission facilities.

Session Year: 2025-2026 House: Assembly Latest Version Date: 2026-08-30

Current Status: **IN PROGRESS** (2026-09-03: Enrolled and presented to the Governor at 4 p.m.)



“Existing law requires the PUC to identify a diverse and balanced portfolio of resources needed to ensure a reliable electricity supply that provides optimal integration of renewable energy and resource diversity in a cost-effective manner, as specified.

This bill would require that the portfolio provide optimal integration in a cost-effective **and risk-prudent manner**, as specified.

New California Legislation: AB 2111

“(2) In developing the guidance pursuant to paragraph (1) the commission shall do all of the following:

(A) Provide an allocation of the renewable energy and zero-carbon resources by region based on consideration of technical feasibility and commercial interest in each region to allow the Independent System Operator to identify and approve transmission facilities needed to interconnect resources and reliably serve the needs of load centers.

(B) Use methods that enable decisionmaking under uncertainty, such as robust optimization or stochastic optimization, to develop candidate near-term planning decisions.

(C) Evaluate candidate planning decisions via a stress testing process through which the cost and reliability of the electrical system and the implications for future planning decisions and flexibility to support the resource portfolios are assessed, and, in a new or existing proceeding, **examine the costs and risks of each candidate planning decision.**

New approaches to change scaling laws

Decomposition & Parallelization

GPU-based Solvers

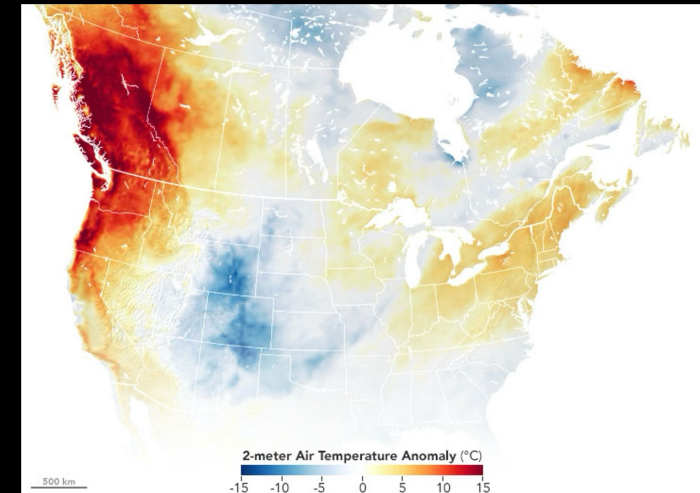
Neural Networks

A unified vision for risk-aware integrated planning

**Deep Uncertainty:
Adaptive Robust
Optimization**

+

**Stochastic Reliability:
Decomposition &
Constraint Generation**



Recall the objective

Determine a least-cost set of actions across several decades that minimizes cost and meets reliability constraints in specific scenarios.

Determine near-term actions that must be taken now that are reasonably affordable and reliable across any plausible future.



Jesse D. Jenkins

jessejenkins@princeton.edu | Twitter: [@JesseJenkins](https://twitter.com/JesseJenkins) | LinkedIn: [jessedjenkins](https://www.linkedin.com/in/jessedjenkins) | Google scholar: <http://bit.ly/ScholarJenkins>